



Alberta Public Health Disease Management Guidelines

Measles

This publication is issued under the Open Government License – Alberta (<http://open.alberta.ca/licence>).

Please note that the terms of this license do not apply to any third-party materials included in this publication.

This publication is available online at <https://open.alberta.ca/publications/measles>

For further information on the use of this guideline contact:

Health.CD@gov.ab.ca

Communicable Disease Control Branch

Public Health Division

Alberta Health

Measles | Alberta Health, Government of Alberta

© 2025 Government of Alberta | May 2025

Table of Contents

Revisions	4
Case Definition	5
Confirmed Case	5
Probable Case	5
Suspect Case	5
Reporting Requirements	6
Physicians, Health Practitioners and Others	6
Laboratories	6
Alberta Health Services and First Nations and Inuit Health Branch	6
Clinical Assessment and Epidemiology	7
Etiology	7
Clinical Presentation.....	7
Diagnosis	7
Treatment.....	7
Reservoir.....	7
Transmission.....	8
Incubation Period	8
Period of Communicability.....	8
Host Susceptibility	8
Incidence.....	8
Public Health Management.....	9
Key Investigation.....	9
Definition of Contact.....	9
Management of a Case.....	10
Management of Contacts	10
Management of Susceptible Contacts.....	11
Outbreak Management.....	14
Preventive Measures.....	14
References	18

Revisions

Revision Date	Document Section	Description of Revision
May 2025	Reporting Requirements	Change to reporting requirement for suspect cases
March 2025	Case Definition	- Updated wording to include exposure in local areas of disease activity - Updated suspect case definition
	Reporting Requirements	Added Fastest Means Possible reporting for suspect cases
July 2024	General	Entire guideline reviewed and updated to align with current information
	Clinical Assessment and Epidemiology	- Updated Diagnosis section to include role of APL - Added "Airborne transmission" - Updated Incidence section to include link to IHDA
	Public Health Management	Updated guidance for managing contacts of measles cases that are communicable during air travel

Case Definition

Confirmed Case

Laboratory confirmation of infection in the absence of recent immunization ^(A) with measles-containing vaccine **AND**:

- Molecular detection of measles virus or isolation by culture from an appropriate clinical specimen, ^(B)
or
- Positive serologic test for measles Immunoglobulin M (IgM) antibody in a person who is either epidemiologically linked to a laboratory-confirmed case or has recent exposure to an area of known measles activity, ^(C)
or
- Seroconversion or a significant rise (e.g., fourfold or greater) in measles Immunoglobulin G (IgG) titre by any standard serologic assay between acute and convalescent sera,

OR

- Clinical illness ^(D) in a person who is epidemiologically linked to a laboratory-confirmed case of measles.

Probable Case

In the absence of both recent immunization ^(A) with a measles-containing vaccine and laboratory confirmation of disease:

- Clinical illness ^(D) in a person with either an epidemiologic link to a non-laboratory-confirmed case of measles or has recent exposure to an area of known measles activity ^(C).

Suspect Case

A person for whom public health or a medical practitioner has a suspicion, based on symptoms and/or exposures, that they may be a measles case, but for whom there is not yet enough information to either classify as a confirmed or probable case or to rule out as a measles case.

^(A) Immunization within 28 days prior to onset of rash or illness.

^(B) Refer to the [Guide to Services](#) for specimen collection information

^(C) See [Measles - WHO](#) for Global incidence of measles; however, recent local epidemiology may not yet be reflected in information presented. [Public Health Notices - Canada](#) may provide recent information for measles outbreaks occurring in Canada.

^(D) Clinical illness, evaluated by a health care professional including public health, includes ALL of the following:

- Fever 38.3°C or greater, AND
- Cough, coryza or conjunctivitis, AND
- Generalized maculopapular rash for at least 3 days.

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed, probable, and suspect cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results:

- by FMP to the MOH (or designate) of the zone, and
- by mail, fax or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed and probable cases.
- The MOH (or designate) shall report all outbreaks to CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or fax).
- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country cases/contacts, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (if applicable),
 - travel history (if known)
 - immunization history (if known), and
 - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Measles is caused by the measles virus, a member of the Paramyxoviridae family, genus *Morbillivirus*.⁽¹⁾

Clinical Presentation

Measles is an acute, highly contagious viral disease that is characterized by some or all of the following prodrome signs and symptoms:

- fever ≥ 38.3 degree Celsius (oral),
- cough, runny nose (coryza) or red eyes (conjunctivitis)
- Koplik spots (clustered blue-white lesions on the buccal mucosa)

The prodrome is followed by:

- a generalized maculopapular rash 3–7 days after prodrome symptom onset and can last for 3–7 days. Rash often begins on face before spread to rest of the body.^(1,2)

Approximately 30% of reported measles cases have one or more complications and are most often seen in children under 5 years of age and those pregnant or immune suppressed.⁽³⁾ Complications of measles infection may include pneumonia, otitis media, febrile seizures, croup, diarrhea, and encephalitis.⁽⁴⁾ Disease in immunocompromised individuals may be more severe with a prolonged course, possibly without the typical rash.⁽⁵⁾

In developed countries, the case-fatality rate is estimated to be less than 1%. The most common causes of measles-related deaths are acute encephalitis in adults and pneumonia in children. In addition, subacute sclerosing panencephalitis (SSPE), a rare, fatal degenerative central nervous system disease may occur many years later in hosts with a history (often before two years of age) of primary measles infection.⁽⁵⁾

Measles infection during pregnancy results in a higher risk of fetal mortality and neurological complications.⁽⁶⁾

Diagnosis

Measles is usually diagnosed based on clinical presentation, exposure history, and laboratory testing.

If an epidemiological link to an already laboratory confirmed case has been established, laboratory testing is not necessary to meet case definition. However, it is extremely valuable to have genotyping information on clusters/outbreaks to support public health surveillance and so testing should be considered. Genotyping helps determine the probable origin of the case and can distinguish between vaccine strains and wild-type virus.

NOTE: Alberta Precision Laboratory (APL) genotypes positive measles specimens to determine if vaccine related. Samples that are identified as wild type are referred to the National Microbiology Laboratory (NML) for genotyping and surveillance.^(personal communication, K. Fonseca, APL, June 22, 2023)

Refer to the [APL Test Directory](#) for information on testing and specimen collection.

Treatment

No specific treatment available. Supportive therapy as indicated.⁽¹⁾

Reservoir

Humans.⁽¹⁾

Transmission

The measles virus is highly contagious and is spread by the airborne and droplet route or by direct contact with respiratory secretions of an infected person, and less commonly, by articles freshly soiled with respiratory secretions.⁽¹⁾ The virus can survive in the air for up to two hours in closed areas.⁽⁵⁾

The secondary attack rate is greater than 90% for susceptible persons.⁽²⁾

Incubation Period

Time from exposure to first signs of disease is approximately 10 days with a range of 7–18 days.⁽²⁾ The incubation period from exposure to rash onset averages 14 days but the rash can appear as late as 19–21 days from infection.^(1,2) Individuals who receive immunoglobulin (Ig) for post-exposure prophylaxis (PEP) may have a prolonged incubation period if they develop disease despite PEP.^(1,2,7)

Period of Communicability

People with confirmed measles are infectious from 4 days prior to onset of the rash to 4 days after the appearance of rash.⁽²⁾ Individuals who are immunocompromised may have prolonged excretion of the virus in respiratory secretions and can be contagious for the duration of their illness.⁽⁷⁾

Host Susceptibility

All persons who have not had the disease nor received age-appropriate immunization are susceptible to infection. Adults born before 1970 are likely to have acquired natural immunity to measles and are generally considered immune. Individuals who recover from measles infection have lifelong immunity.^(1,8)

Infants born to persons who had measles infection are protected against disease for at least the first six to nine months or more due to the passive transfer of antibodies in utero. Children born to persons with vaccine-induced immunity receive less passive antibodies and may be susceptible at an earlier age.⁽¹⁾

Incidence

Even though a safe and cost-effective vaccine is available, in 2021, there were an estimated 128,000 measles deaths globally, mostly among unvaccinated or under vaccinated children under the age of 5 years.⁽⁹⁾ For information on global measles cases see updated [WHO data](#).

Measles became reportable in Canada in 1924. In the pre-vaccine era from 1924 to 1958, an average of 45,000 cases were reported annually. Although measles was eliminated in Canada in 1998, sporadic cases and outbreaks continue post-elimination as a result of importations.⁽¹⁰⁾ Measles immunity in Canada is below the 95% herd immunity threshold required to sustain measles elimination. Therefore, the risk of domestic transmission following an importation of measles remains a reality.⁽¹¹⁾ For information on national measles cases see [Government of Canada Measles and Rubella Weekly Monitoring Reports](#).

Measles became reportable in Alberta in 1985. Measles outbreaks occurred in 2000, 2013 and 2014. In each instance, the initial cases were all travel-related. Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information.

Public Health Management

Key Investigation

- Confirm that the individual meets the case definition.
- Obtain history of illness, including onset date of signs and symptoms and determine period of communicability.
- Determine the occupation of the case e.g., Health Care Worker (HCW). ^(E)
- Determine possible transmission settings (e.g., childcare settings, homeless shelters, overcrowded housing, and health care settings) during the period of communicability.
- Determine measles specific immunization history including:
 - number of doses,
 - date administered,
 - where person was immunized (e.g., out-of-country), and
 - if not immunized, determine reason why.
- Determine the possible source of infection including:
 - recent contact with a confirmed, probable, or suspect case,
 - recent travel history (during the incubation period),
 - recent immigration from an area with known [endemic disease](#), and
 - members of the same household with similar symptoms
- Facilitate collection of all appropriate specimens.
 - Blood, urine, and nasopharyngeal (NP) swab collection is recommended for all potential measles cases.
NOTE: many laboratory collection sites do not collect NP swabs
 - See [Laboratory Testing for Measles](#) for details on timing of samples.
- Identify [contacts](#) that have had an exposure to the case during the period of communicability and assess their immunization history.
 - Risk of transmission is directly related to the proximity and duration of contact and intensity of the exposure.
- Flights: consult MOH for guidance. Refer to [Management of Contacts](#) for more information.

Table 1: Definition of Contact

Definition of Contact
Any persons who have had an exposure to the case during the period of communicability such as; • living in the household with the case, or • had face-to-face contact with the case, or • shared confined air space or was in the same confined air space as the case within a two-hour period after the case had left (e.g., doctor's offices, laboratories, classrooms). There is no minimum time that the case or contact must have been present in the room, or • attends the same school or facility as a case, including all students, staff and volunteers.

^(E) Refer to the [Alberta Immunization Policy \(AIP\)](#) for immunization recommendations and [HCW eligibility](#).

Management of a Case

- Provide information about disease transmission and measures to minimize transmission.
- Hospitalized cases should be managed in accordance with hospital infection prevention and control guidelines.
- Isolation of Confirmed and Probable Cases:
 - The MOH **shall** isolate cases from all public settings, for example, they cannot attend childcare, school or work outside of the home until determined to no longer be infectious.
- Isolation of Suspect Case:
 - The MOH **may** also isolate suspect cases (including those that refuse testing or testing is not done) from all public settings, e.g., they cannot attend childcare, school, or work outside of the home, during the period of communicability if there is a high suspicion of measles disease.

Management of Contacts

- Confirm exposure history and the time since last exposure to the case.
 - The zone MOH assesses the exposure risks in open settings (e.g., retail facilities such as pharmacies or grocery stores) where cases were present while infectious. This may include, but is not limited to, estimations of the:
 - number of susceptible contacts,
 - presence of high-risk individuals (i.e., immunocompromised, pregnant, infant),
 - type of exposure (e.g., proximity to case, duration of exposure), and
 - public health resources available.
- Assess for signs and symptoms,
 - Facilitate [specimen collection](#) as necessary, mindful of infectious period and appropriate infection prevention and control precautions.
- Determine immunization history (i.e., type of vaccine, number of doses and date of administration), including presence of high risk conditions and offer immunization as per the [A/P](#) recommendations.
- Assess for **susceptible contacts** (see [Figure 1: Assessing Measles Contact Susceptibility](#)),
 - The MOH shall exclude susceptible contacts of probable and confirmed cases from all public settings e.g., they cannot attend childcare, school or work outside of the home.⁽¹²⁾ The MOH **may** also exclude susceptible contacts of suspect cases if there is a high suspicion of measles.
 - susceptible contacts are excluded from the 5th day after the first exposure (day zero) to the 21st day after the last exposure, or until they demonstrate an acceptably low risk for infecting others (through serology and/or post-exposure prophylaxis (PEP)).
 - Identify high-risk susceptible contacts i.e., considered at high-risk for measles complications:
 - immunocompromised individuals with conditions listed in the [Canadian Immunization Guide \(CIG\)](#),
 - pregnant persons, and
 - infants.
- For management of contacts of cases that were communicable during air travel, the Public Health Agency of Canada (PHAC) recommends a risk-based approach to contact-tracing.
 - Providing notification of a measles exposure within the past 21 days using public media is beneficial to raise awareness and provide public health recommendations to all flight and non-flight related potential contacts.
 - Requesting a flight manifest for contact management is generally not feasible because case identification and reporting often occurs too late for public health to notify individuals and assess/arrange for PEP within the 72 hour treatment window for vaccine or 6 days for immunoglobulin.
 - However, a flight manifest can be requested at the discretion of the MOH in unusual circumstances, such as when a case is reported to public health shortly after flight arrival and a flight manifest is available in a timely manner.
 - See [Table 2: Considerations for Flight Manifest Request](#)

- For more information about contact management for measles cases communicable during air travel, see Figure 2: Measles on a Flight – Contact Tracing. (13)

Table 2: Considerations for Flight Manifest Request⁽¹³⁾

Considerations

- Rarely, will diagnosis of a measles case be in sufficient time that a flight manifest is useful for individual passenger contact for purposes of PEP.
- Flight manifests often lack relevant/current contact information (e.g., emails).
- The turnaround time required to secure and receive the manifest can be lengthy.

Management of Susceptible Contacts

Determine if PEP and/or exclusion are required for susceptible contacts. ([see Figure 1](#))

While the PEP options are the same for all Albertans, the steps required in the overall management of susceptible contacts (serology, PEP, exclusions) differ for the following:

- infants < 6 months of age (see [Figure 3: Post-Exposure Prophylaxis and Exclusion Recommendations for Infants < 6 Months of Age](#)),
- non- health care workers (see [Figure 4: Post-Exposure Management of Susceptible Members of the Public \(Non-HCWs\)](#)),
- [health care workers](#), (see [Figure 5: Post-Exposure Management of Susceptible Health Care Workers](#) (HCW)).

Post-Exposure Prophylaxis (PEP)

PEP given to susceptible contacts in the appropriate timeframe can modify or prevent disease.^(10,14) Susceptible contacts of a measles case may be eligible to receive either MMR vaccine or immunoglobulin (Ig) depending on the time-lapse from exposure, age, and health status.^(8,15) See the [AIP](#) for reference on the recommendations for vaccine and Ig.

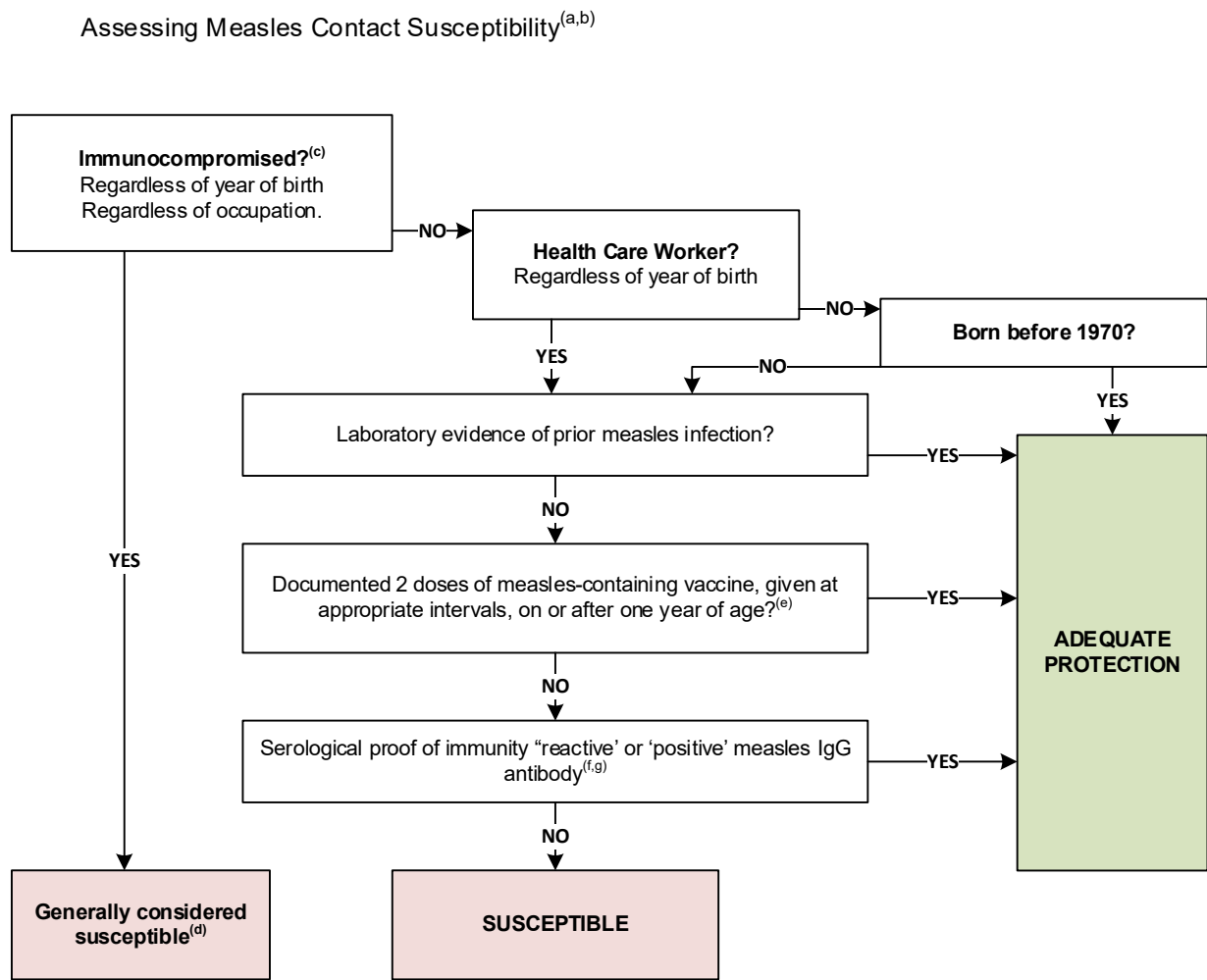
Vaccine

- Immunization should be the primary prophylactic intervention for all susceptible immunocompetent contacts six months of age and older, unless measles containing vaccine is contraindicated. ^(8,15)
NOTE: Do not delay administration of vaccine beyond 72 hours pending serology results. Vaccine given beyond 72 hours does not offer protection for the current exposure, but it does protect against subsequent exposures.

Immunoglobulin (Ig)

- Ig should not be used to control outbreaks and is not generally indicated for contacts who have received one dose of vaccine at 12 months of age or older.
- Consider Ig administration for the following susceptible contacts:
 - [High-risk individuals](#) for whom vaccine is contraindicated, particularly:
 - immunocompromised people,
 - pregnant persons, and
 - infants 6–12 months of age who cannot receive MMR vaccine within 72 hours of exposure.^(8,15)
 - All infants < 6 months of age.
 - HIV Infected Persons: Measles antibody titre is known to decline more rapidly over time in HIV-infected children as compared to HIV-uninfected children.⁽⁸⁾
 - Others may be considered, on a case-by-case basis in consultation with the CMOH.
- Offering Ig beyond six days of first exposure is not recommended. It does not prevent disease and delays immunization to protect from further exposures.
- See the [AIP](#) for dosage information.

Figure 1: Assessing Measles Contact Susceptibility



^(a) For susceptibility and management of infants under 6 months of age. see [Figure 3](#)

^(b) When determining who is adequately protected, consideration should be given to individuals who have recently received Ig or intravenous immune globulin (IVIG). Duration of protection will depend on the dose, frequency of administration of IVIG and immunocompetence. Consultation with a specialist may be required.

^(c) Refer to [Immunization of Specific Populations in the AIP](#) for further information. Consultation with a specialist may be required.

^(d) See [Figure 5](#) for management of immunocompromised health care workers.

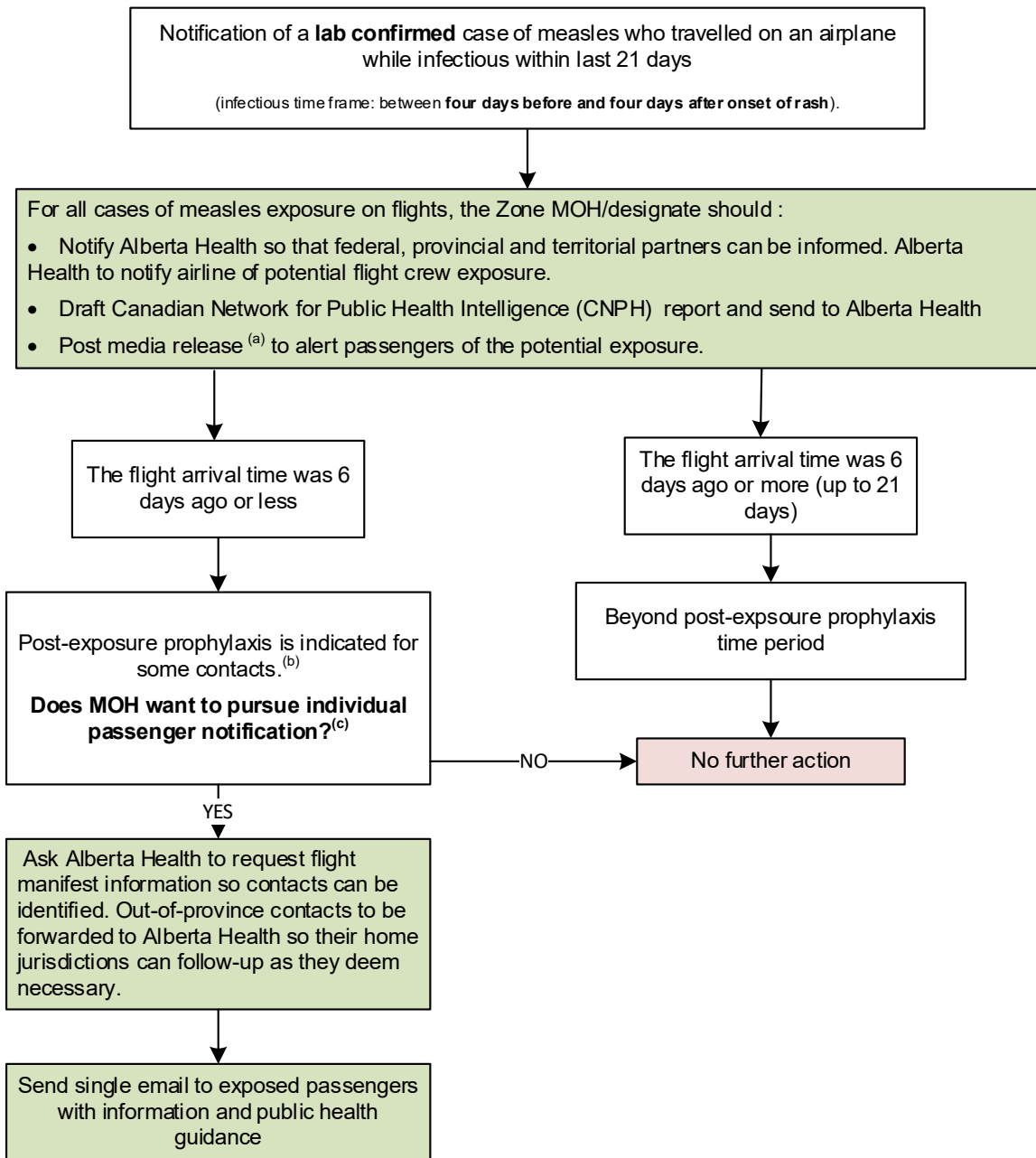
^(e) Children who are up-to-date for age (1 year of age up to and including 3 years of age, with one dose of vaccine) are considered susceptible and should be managed accordingly.

^(f) There is no expiry on IgG positive results. However, all individuals born in or after 1970 who are IgG positive following one dose of vaccine, should receive a second dose of vaccine to ensure comprehensive protection.

^(g) Serological proof of immunity should generally be considered for individuals 12 months of age or older. However, serology may also be considered for infants 6-11 months of age with a history of immunization. Note: IgG serology is not a reliable measure of immunity for infants whose only protection comes from maternal antibodies, i.e., unimmunized.

Note: Individuals who have serology completed >7 days after exposure should be confirmed to be asymptomatic before being considered immune.

Figure 2: Measles on a Flight – contact tracing⁽¹³⁾



^(a)Public messaging should contain flight details, dates of exposure, and any additional community exposures, and instructions for monitoring for symptoms and seeking care.

^(b) Refer to [Figure 3](#), [Figure 4](#) and [Figure 5](#) as appropriate.

^(c) See [Table 2: Considerations for pursuing individual passenger notification](#).

Long-Term Protection

All contacts initially identified as susceptible in an investigation should be evaluated for long-term measles protection according to recommendations in the [AIP](#). This includes individuals who receive vaccine or Ig as PEP and who may then be considered protected for the current exposure.

Outbreak Management

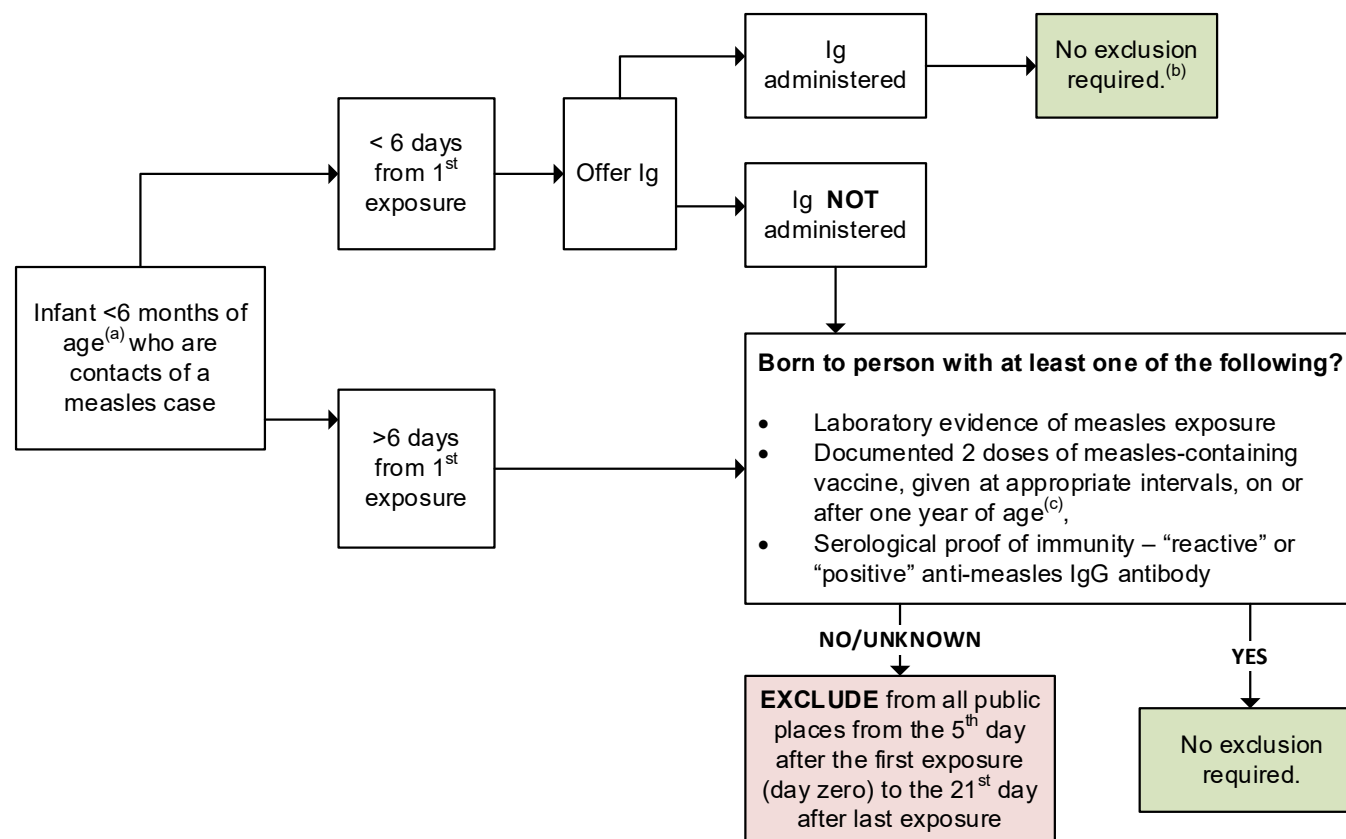
Outbreak definition:

- Two or more confirmed cases of measles, linked either epidemiologically and/or virologically.⁽¹²⁾
- During a measles outbreak, the MOH may recommend either or both of the following, in consultation with the CMOH to ensure continuity of vaccine supply:
 - Immunizing children 6-11 months of age inclusive, with MMR vaccine,
 - Offering an early second dose of measles-containing vaccine, respecting the minimum interval between doses. Minimum intervals vary depending on the vaccine product. See the [AIP](#) for details.

Preventive Measures

- Public education about the risks of measles disease and the importance of immunization.
- Immunization of all eligible Albertans according to recommendations in the current [AIP](#).
 - Assessment of protection is particularly important for the following groups:
 - All [health care workers](#)
 - Other adults who are at greatest risk of exposure, regardless of year of birth (e.g., students at post-secondary institutions, military recruits),
 - Individuals travelling to countries with circulating measles disease, and
 - All adults born in 1970 or later.
- Remind individuals planning on taking a trip to seek travel advice weeks prior to leaving, especially if going to a location with known measles cases/outbreak(s). Refer to the Government of Canada's [Travel Advice and Advisories by destination](#) website for travel outside of Canada or refer to the [AIP](#) for recommendations if traveling within Canada.

Figure 3: Post-Exposure Prophylaxis and Exclusion Recommendations for Infants < 6 Months of Age ^(12,16)

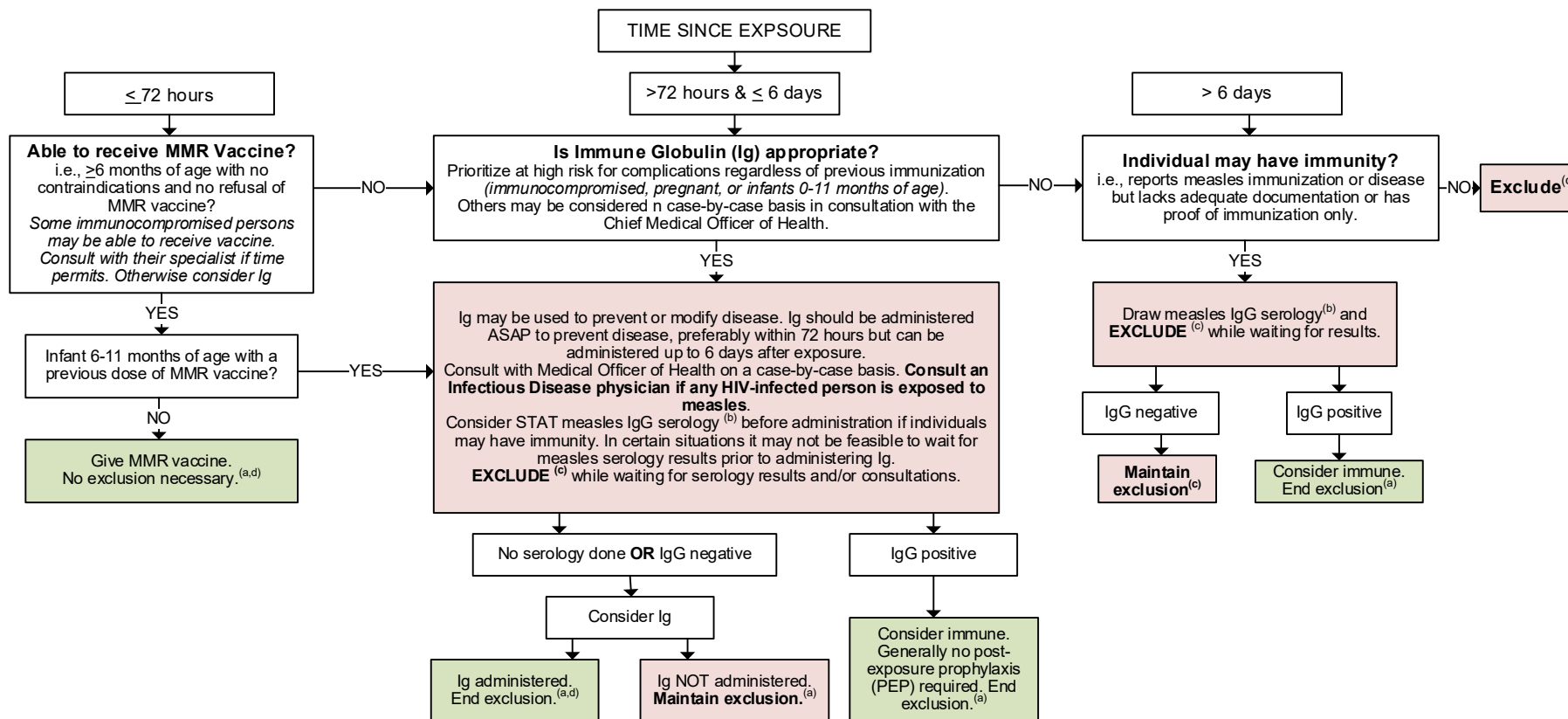


^(a) The susceptibility of infants <6 months of age is determined by maternal immunity. Infants born to those without evidence of immunity are considered susceptible.⁽¹³⁾ Infants born to those with evidence of immunity are generally considered to be immune due to passive antibody transfer, therefore exclusions are not required. However, Ig is recommended as a precaution for this high-risk group due to potential for waning maternal antibodies.

^(b) Although infants who are born to those with immunity may have some protection through maternal antibodies, Ig should be offered to all contacts of measles case under 6 months of age if time permits.

^(c) There is no expiry on IgG positive results.

Figure 4: Post-Exposure Management of Susceptible Members of the Public (Non-HCWs)⁽¹⁷⁾



^(a) Counselling regarding signs and symptoms of measles disease and self-reporting is recommended for all contacts, regardless of serology results and administration of post-exposure prophylaxis.

^(b) Positive IgG serology in the following asymptomatic susceptible contact can be considered proof of immunity:

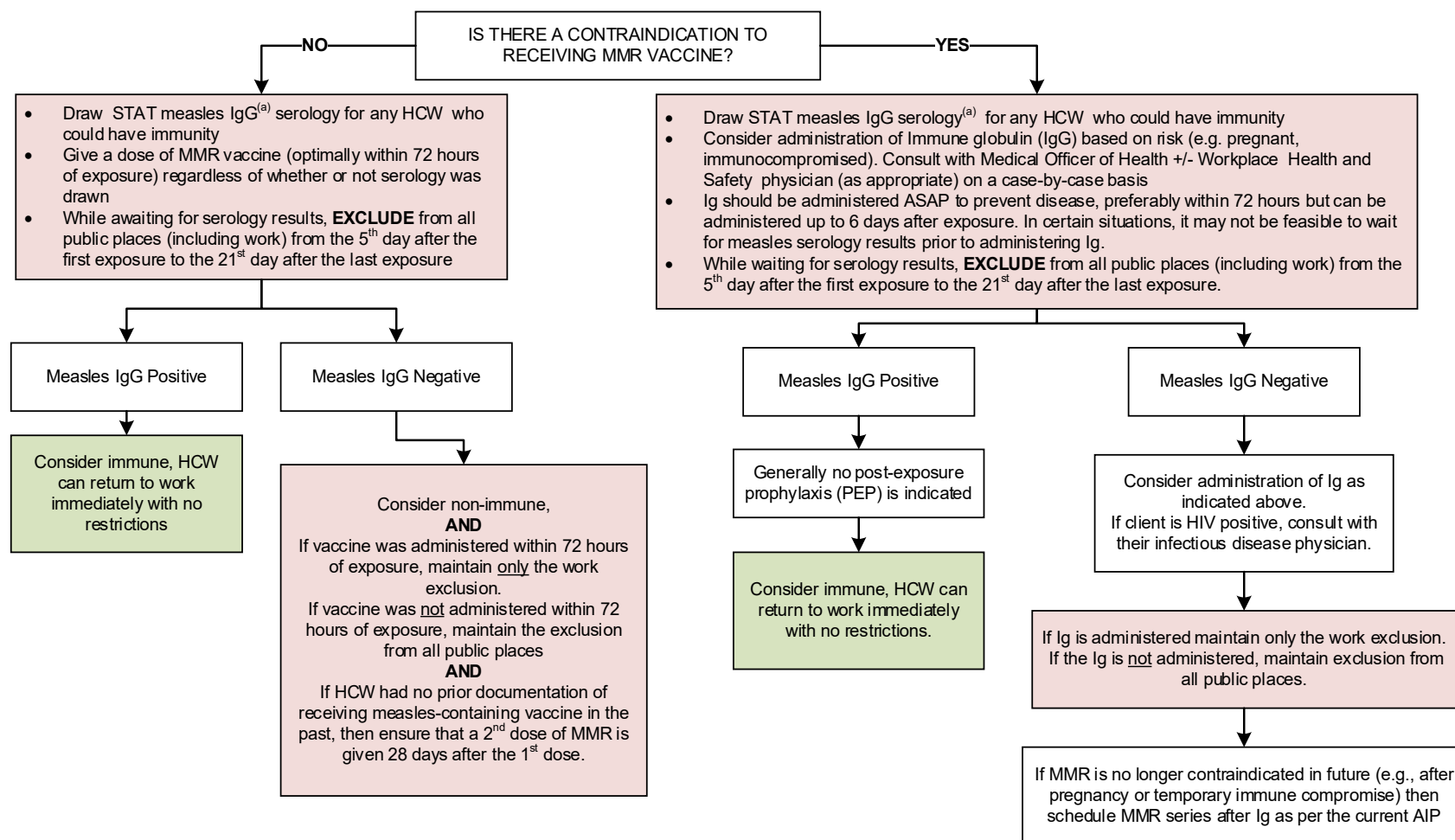
- 6-11 months of age and received a dose of measles containing vaccine ≥ 14 days before exposure.
- Individuals who have serology completed >7 days after exposure should be confirmed to be asymptomatic before being considered immune.

NOTE: Do not draw measles IgG serology for infants under 6 months of age, or for infants 6-12 months of age whose only protection is from maternal antibodies (i.e., unimmunized), because a positive result may not be reliable.

^(c) Exclude from all public places from the 5th day after the first exposure to the 21st day after last exposure

^(d) Post-exposure prophylaxis is not 100% effective. Advise clients to avoid unnecessary contact with individuals at high risk for complications until 21 days after their last exposure

Figure 5: Post-Exposure Management of Susceptible Health Care Workers (HCW) ⁽¹²⁾



^(a) A positive IgG results in an asymptomatic contact can be considered proof of immunity

NOTE: Individuals who have serology completed >7 days after exposure should be confirmed to be asymptomatic before being considered immune.

References

1. Strebel, P. Perry R. Measles. In: Heymann DL (ed.), editor. Control of communicable Disease Manual. 21st ed. Washington D.C.: American Public Health Association; 2022. p. 405–14.
2. Government of Canada. Measles: For health professionals - Canada.ca [Internet]. [cited 2024 Apr 11]. Available from: <https://www.canada.ca/en/public-health/services/diseases/measles/health-professionals-measles.html>
3. Government of Canada. Measles: For health professionals - Canada.ca [Internet]. 2024 [cited 2024 Apr 25]. Available from: <https://www.canada.ca/en/public-health/services/diseases/measles/health-professionals-measles.html>
4. World Health Organization. Measles [Internet]. [cited 2023 Jul 13]. Available from: <https://www.who.int/news-room/fact-sheets/detail/measles>
5. Hall, E. Wap H. Measles - Vaccine Preventable Diseases Surveillance Manual | CDC [Internet]. [cited 2023 Feb 24]. Available from: <https://www.cdc.gov/vaccines/pubs/surv-manual/chpt07-measles.html>
6. Ragusa R, Platania A, Cuccia M, Zappalà G, Giorgianni G, D'Agati P, et al. Measles and Pregnancy: Immunity and Immunization—What Can Be Learned from Observing Complications during an Epidemic Year. J Pregnancy [Internet]. Hindawi Limited; 2020 [cited 2023 Jul 13];2020. Available from: [/pmc/articles/PMC7416282/](https://doi.org/10.1155/2020/7416282)
7. American Academy of Pediatrics. Measles. In: Kimberlin DW, editor. Red Book: 2021–2024 Report of the Committee on Infectious Diseases. 32nd ed. Itasca IL: American Academy of Pediatrics; 2021. p. 503–19.
8. Government of Alberta | Alberta Health. Measles-Mumps-Rubella Combined Vaccine Revision Date: May 18, 2022. 2022;(May):1–8.
9. Measles [Internet]. [cited 2024 Mar 4]. Available from: <https://www.who.int/news-room/fact-sheets/detail/measles>
10. Government of Canada. Measles surveillance in Canada: 2016 [Internet]. 2017. Available from: <https://www.canada.ca/en/public-health/services/publications/healthy-living/canadian-immunization-guide-part-4-active-vaccines/page-12-measles-vaccine.html>
11. Public Health Agency of Canada (PHAC). Measles in Canada. Canada Commun Dis Rep [Internet]. 2014;40(12). Available from: www.canada.ca/en/public-health/services/reports-publications/canada-communicable-disease-report-ccdr/monthly-issue/2014-40/ccdr-volume-40-12-june-12-2014/ccdr-volume-40-12-june-12-2014-3.html
12. Public Health Agency of Canada (PHAC). Guidelines for the prevention and control of measles outbreaks in Canada. Can Commun Dis Rep. 2013. p. 1–52.
13. Public Health Agency of Canada. Process for contact management for measles cases communicable during air travel. 2024.
14. Alberta Health. Alberta Immunization Policy [Internet]. 2017. Available from: www.health.alberta.ca/professionals/immunization-policy.html
15. Public Health Agency of Canada (PHAC). Updated NACI recommendation for measles post-exposure prophylaxis. Canada Commun Dis Rep [Internet]. 2018 Sep 6;44(9):226–30. Available from: www.canada.ca/content/dam/phac-aspc/documents/services/reports-publications/canada-communicable-disease-report-ccdr/monthly-issue/2018-44/issue-9-september-6-2018/ccdrv44i09a07-eng.pdf
16. Measles vaccines: Canadian immunization guide - Canada.ca [Internet]. [cited 2024 Feb 27]. Available from: <https://www.canada.ca/en/public-health/services/publications/healthy-living/canadian-immunization-guide-part-4-active-vaccines/page-12-measles-vaccine.html>
17. National Advisory Committee on Immunization (NACI). Measles Vaccine [Internet]. Canadian Immunization Guide. Public Health Agency of Canada; 2017. Available from: www.canada.ca/en/public-health/services/publications/healthy-living/canadian-immunization-guide-part-4-active-vaccines/page-12-measles-vaccine.html